

Mid Point Circle Algorithm

The equation of a circle is given by

$$x^2 + y^2 = r^2$$

To apply the midpoint method, we define a circle function

as $F_{\text{circle}}(x,y) = x^2 + y^2 - r^2$

now $F_{\text{circle}}(x,y) < 0$ if (x,y) is inside the circle boundary
 $= 0$ if (x,y) is on the circle boundary
 > 0 if (x,y) is outside the circle boundary

This circle function $F_{\text{circle}}(x,y)$ serves as the decision parameter

Select next pixel along the circle path according to the sign of circle function evaluated at the midpoint between two candidate pixels.

Start at $(0,y)$ take unit steps in 'x' direction (sample in 'x' direction $x_{k+1} = x_k + 1$)

Assuming position (x_k, y_k) has been selected at previous step we determine next position

(x_{k+1}, y_{k+1}) as either (x_{k+1}, y_k) or $(x_{k+1}, y_k - 1)$ along circle path by evaluating the decision parameter (circle function). The decision parameter is the circle function evaluated at the midpoint between these two pixels

$$P_k = F_{\text{circle}}(x_k + 1, y_k - \frac{1}{2})$$

$$= (x_k + 1)^2 + (y_k - \frac{1}{2})^2 - r^2 \text{ ----- (i)}$$

At the next sampling position $(x_{k+1} + 1 = x_k + 2)$, the decision parameter is evaluated as

$$P_{k+1} = F_{\text{circle}}(x_{k+1} + 1, y_{k+1} - \frac{1}{2})$$

$$= [(x_k + 1) + 1]^2 + (y_{k+1} - \frac{1}{2})^2 - r^2 \text{ ----- (ii)}$$

Now subtracting eq (i) and (ii),

$$P_{k+1} = P_k + 2(x_k + 1) + (y_{k+1}^2 - y_k^2) - (y_{k+1} - y_k) + 1 \text{ ----- (iii)}$$

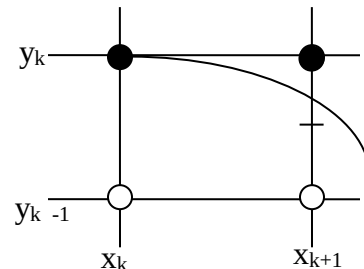
where y_{k+1} is either y_k or $y_k - 1$ depending on the sign of P_k .

Case 1:

If $P_k < 0$ then the mid point is inside the circle, so pixel on scanline ' y_k ' is closer to the circle boundary and $y_{k+1} = y_k$

From equation (iii)

or $P_{k+1} = P_k + 2x_{k+1} + 1 \text{ ----- (a)}$



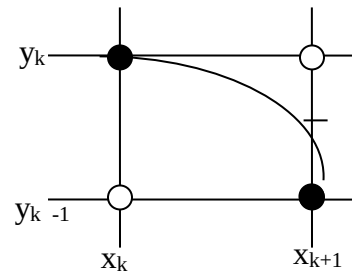
Where $x_{k+1} = x_k + 1$

or $2x_{k+1} = 2x_k + 2$

Case 2:

If $P_k \geq 0$ then the mid point is outside or on the boundary of the circle, so we select the pixel on scan line ' $y_k - 1$ ' then $y_{k+1} = y_k - 1$ i.e. from equation (iii)

$$\text{or } P_{k+1} = P_k + 2x_{k+1} - 2y_{k+1} + 1 \quad \text{----- (b)}$$



$$\text{Where } 2y_{k+1} = 2y_k - 2$$

$$\text{or } 2x_{k+1} = 2x_k + 2$$

The initial decision parameter P_0 is obtained by evaluating the circle function at the starting position $(x_0, y_0) = (0, r)$

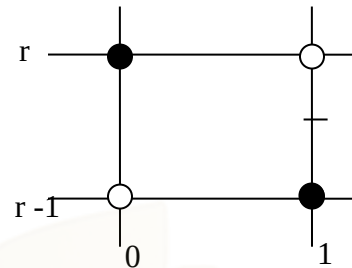
Next pixel to plot is either $(1, r)$ or $(1, r - 1)$

So, midpoint coordinate position is $(1, r - \frac{1}{2})$

$$F_{\text{circle}}(1, r - \frac{1}{2}) = 1 + (r - \frac{1}{2})^2 - r^2$$

Thus,

$$P_0 = 5/4 - r$$



If the radius ' r ' is specified as an integer, we can simply round P_0 to $P_0 = 1 - r$ (for ' r ' an integer)

Midpoint Circle Algorithm

1. Input radius r and circle center (x_c, y_c) , and obtain the first point on the circumference of a circle centered on the origin as $(x_0, y_0) = (0, r)$

2. Calculate the initial value of the decision parameter as

$$P_0 = 5/4 - r$$

3. At each x_k position, starting at $k = 0$, perform the following test:

If $P_k < 0$, the next point along the circle centered on $(0,0)$ is (x_{k+1}, y_k) and $P_{k+1} = P_k + 2x_{k+1} + 1$

Otherwise, the next point along the circle is $(x_k + 1, y_k - 1)$ and $P_{k+1} = P_k + 2x_{k+1} - 2y_{k+1} + 1$

$$\text{where } 2x_{k+1} = 2x_k + 2 \text{ and } 2y_{k+1} = 2y_k - 2.$$

4. Determine symmetry points in the other seven octants.

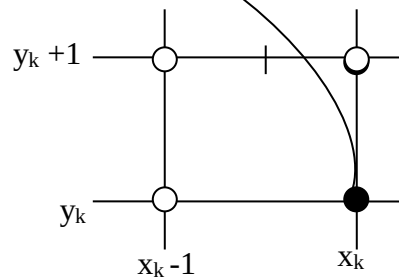
5. Move each calculated pixel position (x, y) onto the circular path centered on (x_c, y_c) and plot the coordinate values:

$$x = x + x_c, \quad y = y + y_c$$

6. Repeat steps 3 through 5 until $x \geq y$.

Midpoint Circle Algorithm

Starting point is at $(r,0)$ and moving in anticlockwise direction



$$P_0 = 1 - r$$

If $P_k < 0$, the next point along the circle centered on $(0,0)$ is (x_k, y_k+1) and

$$P_{k+1} = P_k + 2y_{k+1} + 1$$

Otherwise, the next point along the circle is $(x_k - 1, y_k + 1)$ and

$$P_{k+1} = P_k - 2x_{k+1} + 2y_{k+1} + 1$$

Digitize a circle with a radius of 10 pixels and starting point at $(10,0)$ and moving in anticlockwise direction

$$P_0 = 1 - 10 = -9$$

K	P_k	x_{k+1}	y_{k+1}
0	$P_0 = -9$	10	1
1	$P_1 = -9 + 2 + 1 = -6$	10	2
2	$P_2 = -6 + 4 + 1 = -1$	10	3
3	$P_3 = -1 + 6 + 1 = 6$	9	4
4	$P_4 = 6 - 18 + 8 + 1 = -3$	9	5
5	$P_5 = -3 + 10 + 1 = 7$	8	6
6	$P_6 = 7 - 16 + 12 + 1 = 4$	7	7